EXPERIMENT 3: ε-CLOSURE OF NFA

Aim:

To write a C program to find the ε-closure of all states in a Non-Deterministic Finite Automaton (NFA) with ε-moves.

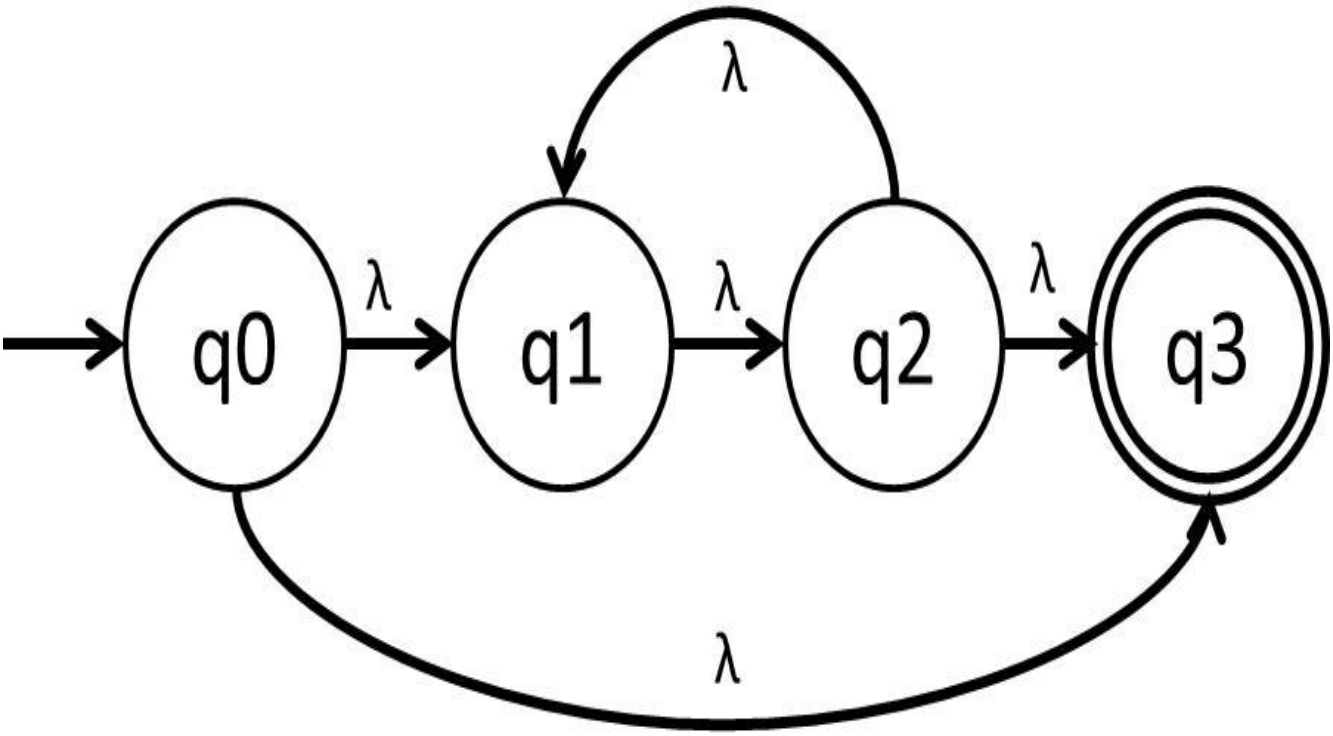
Procedure / Algorithm:

1. Start the program.
2. Read the number of states and ε-transitions.
3. Store the ε-transitions in an adjacency matrix.
4. Include each state in its own ε-closure.
5. Find all states reachable through ε-transitions, including indirect transitions.
6. Display the ε-closure of every state.
7. Stop the program.

Example:

The ε-closure of a state is the set of all states reachable from that state using zero or more ε-transitions, including the state itself. An ε-transition allows the automaton to move between states without consuming an input symbol.

Sample Design:



Transition table:

Present state	Input a	Input \(\varepsilon\)
→ q_0	nil	q_1
q_1	nil	q_2
q_2	q_2	q_3
*q_3	nil	nil

ε-closure table:

State	ε-closure
q_0	q_0,q_1,q_2,q_3
q_1	q_1,q_2,q_3
q_2	q_2,q_3
q_3	q_3

Code:

```c
#include <stdio.h>
int main()
{
    int n, m, i, j, a, b;
    int e[10][10] = {0};
    printf("Enter number of states: ");
    scanf("%d", &n);
    printf("Enter number of epsilon transitions: ");
    scanf("%d", &m);
    for (i = 0; i < m; i++)
    {
        scanf("%d%d", &a, &b);
        e[a][b] = 1;
    }
    for (i = 0; i < n; i++)
    {
        printf("E-closure(q%d) = { q%d ", i, i);
        for (j = 0; j < n; j++)
            if (e[i][j])
                printf("q%d ", j);
        printf("}\n");
    }
    return 0;
}
```

```
Enter number of states: 3
Enter number of epsilon transitions: 2
1 0
0 1
E-closure(q0) = { q0 q1 }
E-closure(q1) = { q1 q0 }
E-closure(q2) = { q2 }
```

Input / Output

Input:

Enter number of states: 3

Enter number of epsilon transitions: 2
0 1
1 2

Output:

Epsilon closures:
E-closure(q0) = { q0 q1 q2 } E-
closure(q1) = { q1 q2 }
E-closure(q2) = { q2 }

Result

Thus, the C program to find the ε-closure of all states in an NFA with ε-moves was executed successfully.